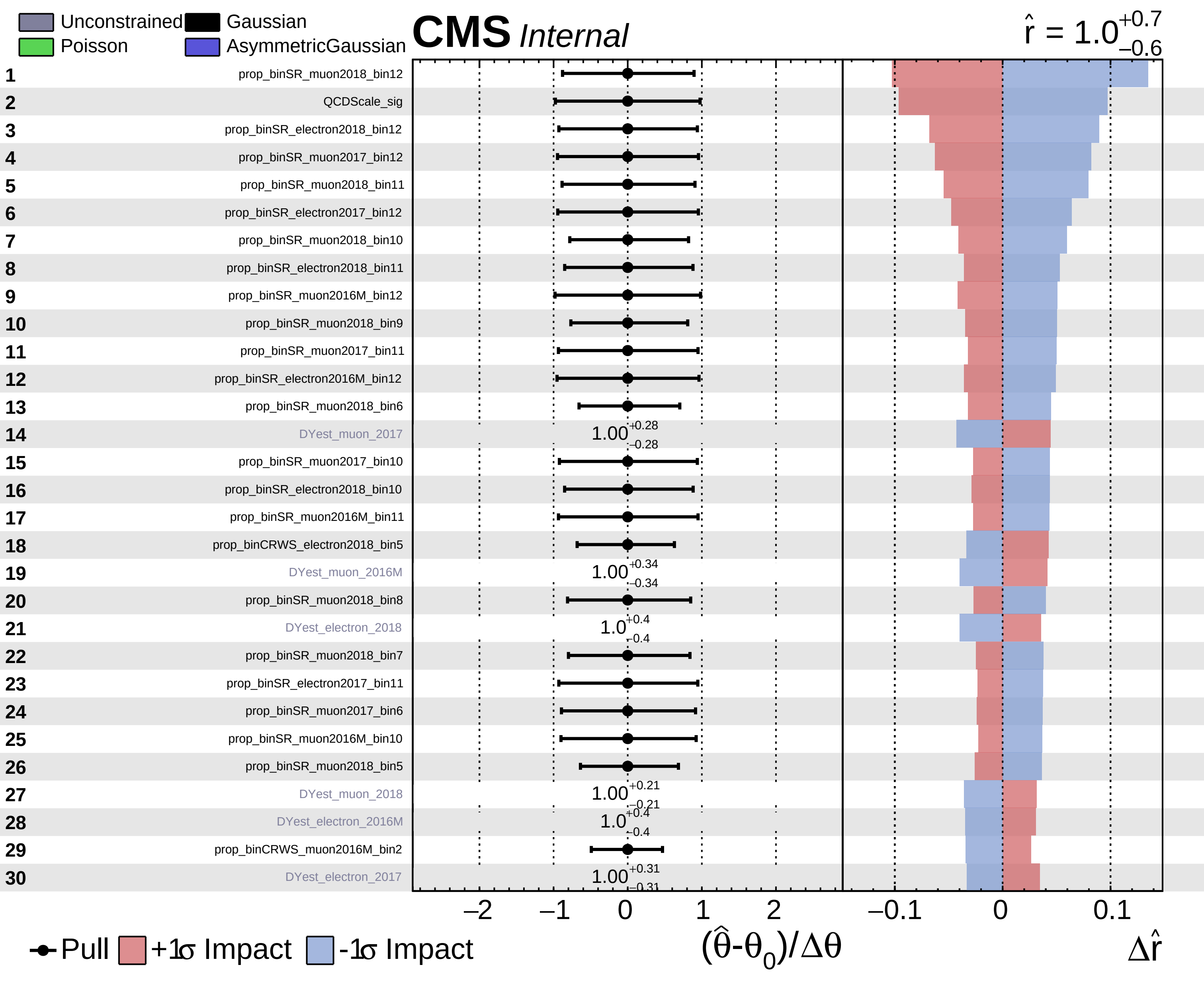
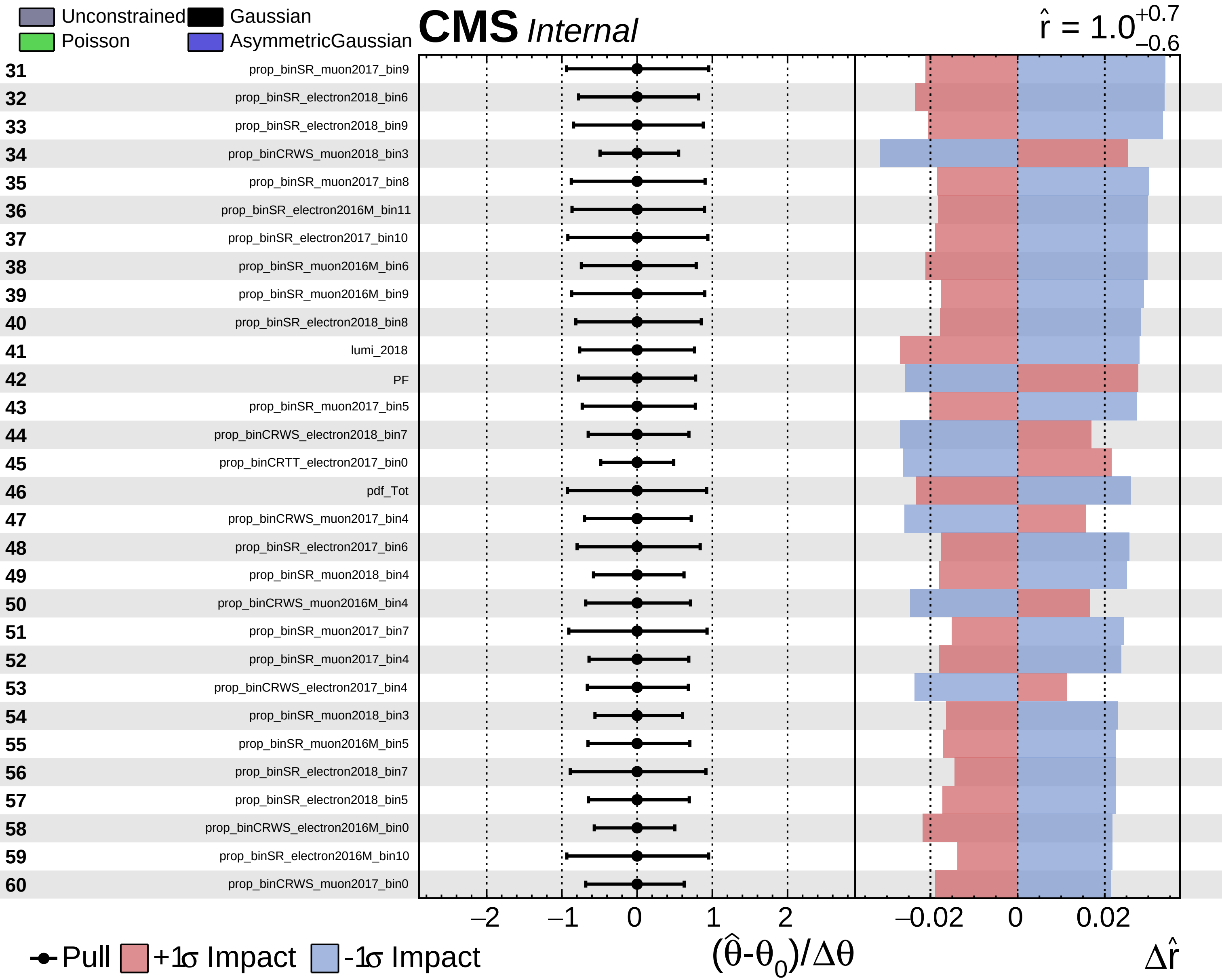
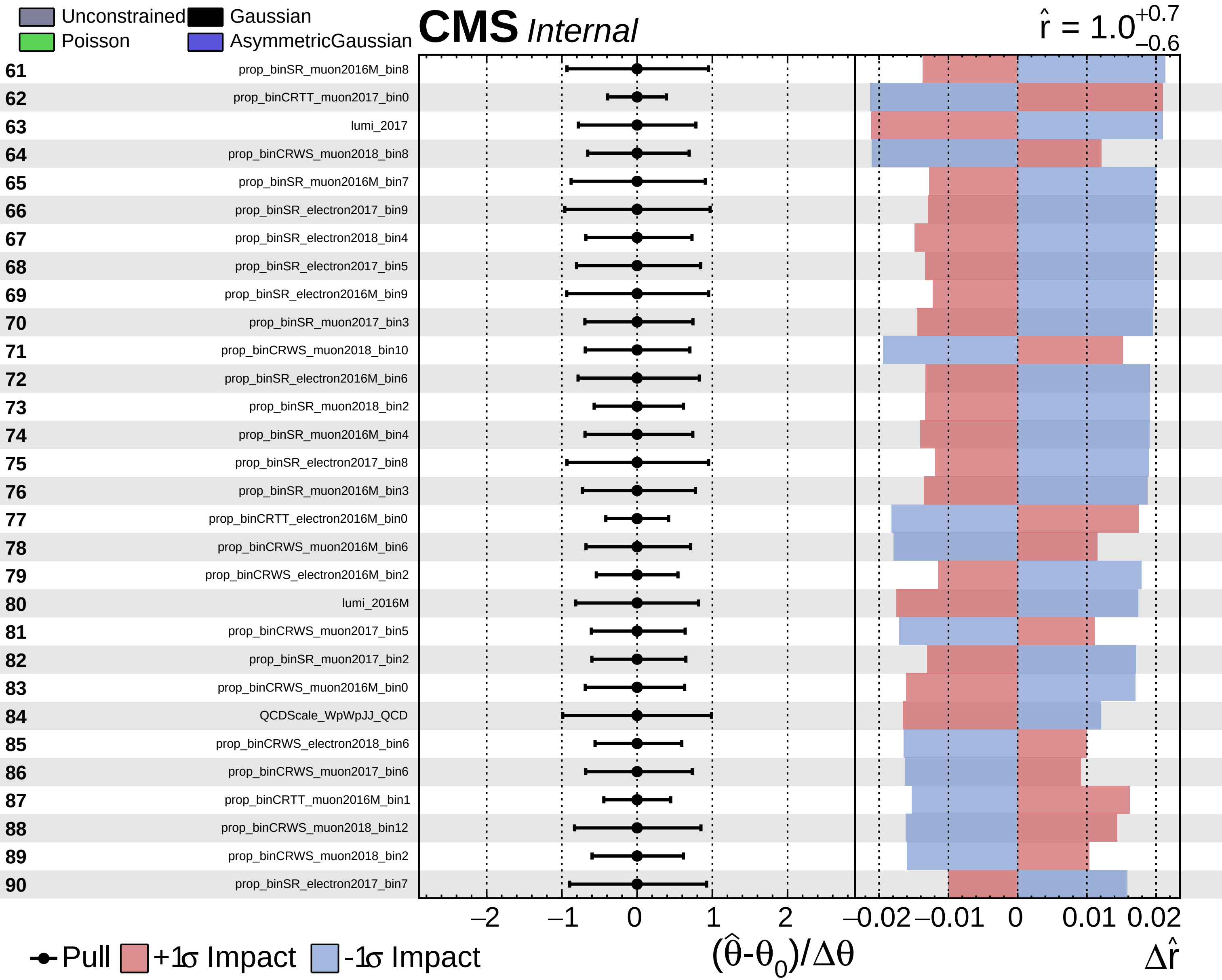


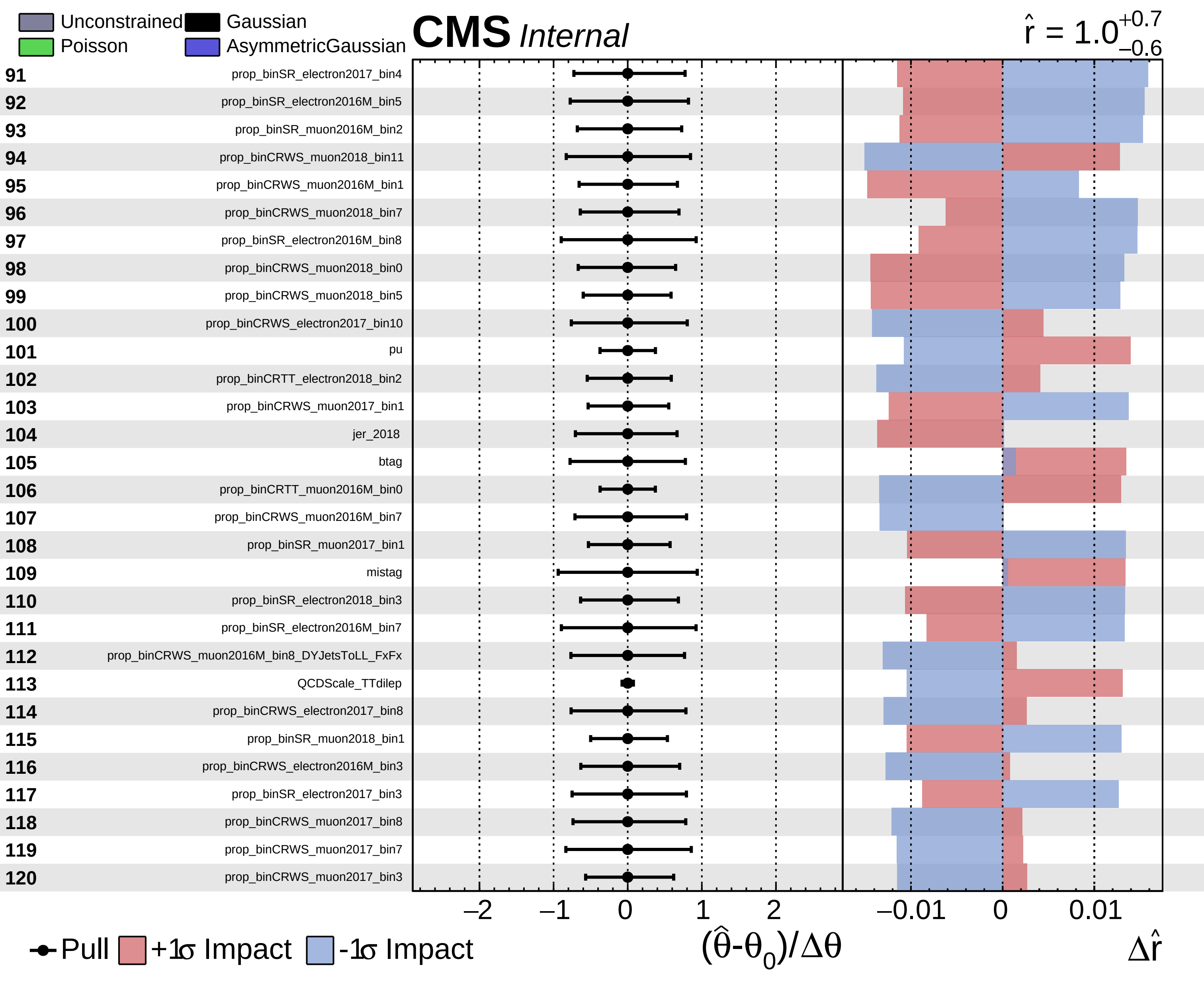
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
1 prop_binSR_muon2018_bin12
2 QCDScale_sig
3 prop_binSR_electron2018_bin12
4 prop_binSR_muon2017_bin12
5 prop_binSR_muon2018_bin11
6 prop_binSR_electron2017_bin12
7 prop_binSR_muon2018_bin10
8 prop_binSR_electron2018_bin11
9 prop_binSR_muon2016M_bin12
10 prop_binSR_muon2018_bin9
11 prop_binSR_muon2017_bin11
12 prop_binSR_electron2016M_bin12
13 prop_binSR_muon2018_bin6
14 DYest_muon_2017 $1.00^{+0.28}_{-0.28}$
15 prop_binSR_muon2017_bin10
16 prop_binSR_electron2018_bin10
17 prop_binSR_muon2016M_bin11
18 prop_binCRWS_electron2018_bin5
19 DYest_muon_2016M $1.00^{+0.34}_{-0.34}$
20 prop_binSR_muon2018_bin8
21 DYest_electron_2018 $1.0^{+0.4}_{-0.4}$
22 prop_binSR_muon2018_bin7
23 prop_binSR_electron2017_bin11
24 prop_binSR_muon2017_bin6
25 prop_binSR_muon2016M_bin10
26 prop_binSR_muon2018_bin5
27 DYest_muon_2018 $1.00^{+0.21}_{-0.21}$
28 DYest_electron_2016M $1.0^{+0.4}_{-0.4}$
29 prop_binCRWS_muon2016M_bin2
30 DYest_electron_2017 $1.00^{+0.31}_{-0.31}$
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



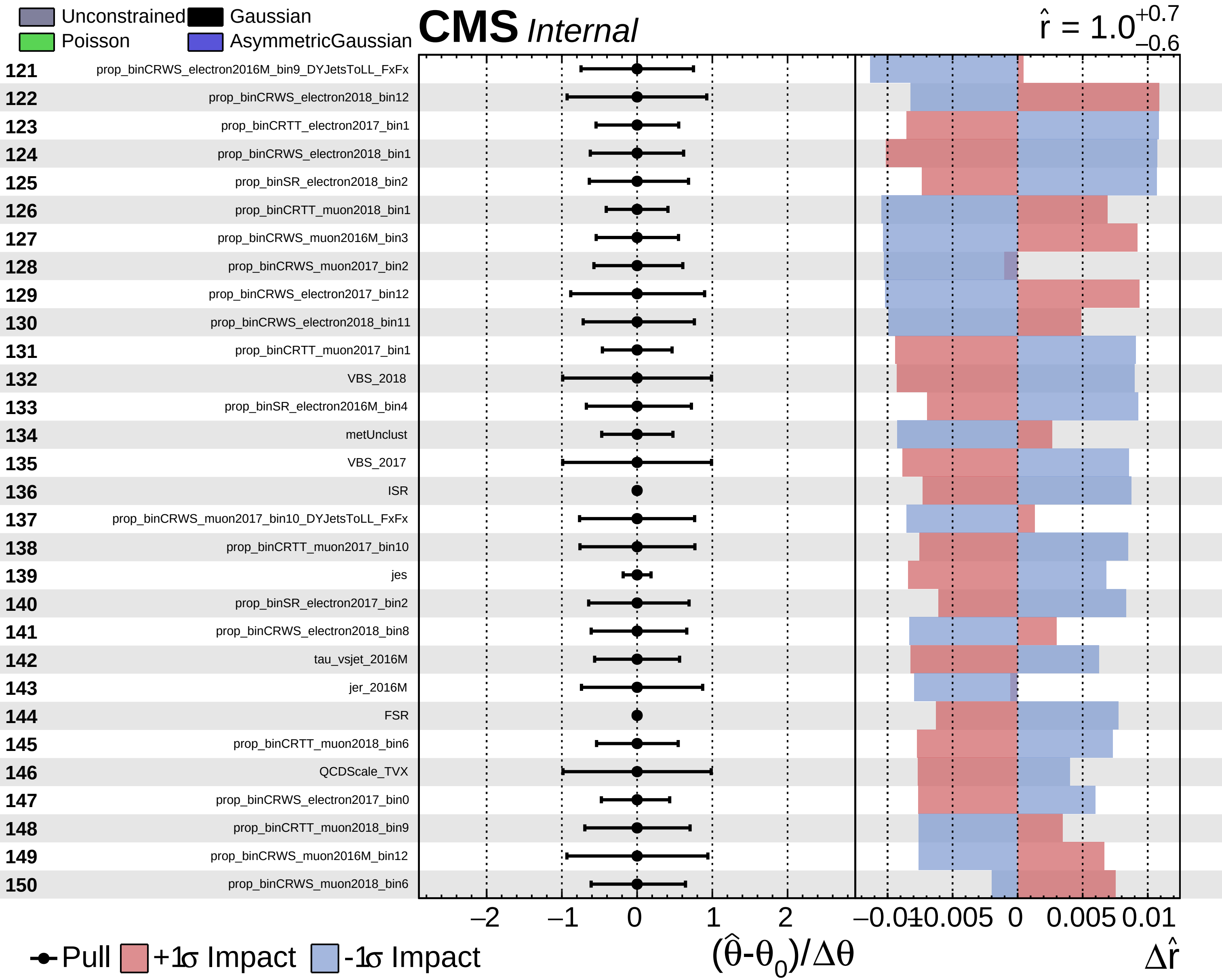
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
31 prop_binSR_muon2017_bin9
32 prop_binSR_electron2018_bin6
33 prop_binSR_electron2018_bin9
34 prop_binCRWS_muon2018_bin3
35 prop_binSR_muon2017_bin8
36 prop_binSR_electron2016M_bin11
37 prop_binSR_electron2017_bin10
38 prop_binSR_muon2016M_bin6
39 prop_binSR_muon2016M_bin9
40 prop_binSR_electron2018_bin8
41 lumi_2018
42 PF
43 prop_binSR_muon2017_bin5
44 prop_binCRWS_electron2018_bin7
45 prop_binCRTT_electron2017_bin0
46 pdf_Tot
47 prop_binCRWS_muon2017_bin4
48 prop_binSR_electron2017_bin6
49 prop_binSR_muon2018_bin4
50 prop_binCRWS_muon2016M_bin4
51 prop_binSR_muon2017_bin7
52 prop_binSR_muon2017_bin4
53 prop_binCRWS_electron2017_bin4
54 prop_binSR_muon2018_bin3
55 prop_binSR_muon2016M_bin5
56 prop_binSR_electron2018_bin7
57 prop_binSR_electron2018_bin5
58 prop_binCRWS_electron2016M_bin0
59 prop_binSR_electron2016M_bin10
60 prop_binCRWS_muon2017_bin0
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

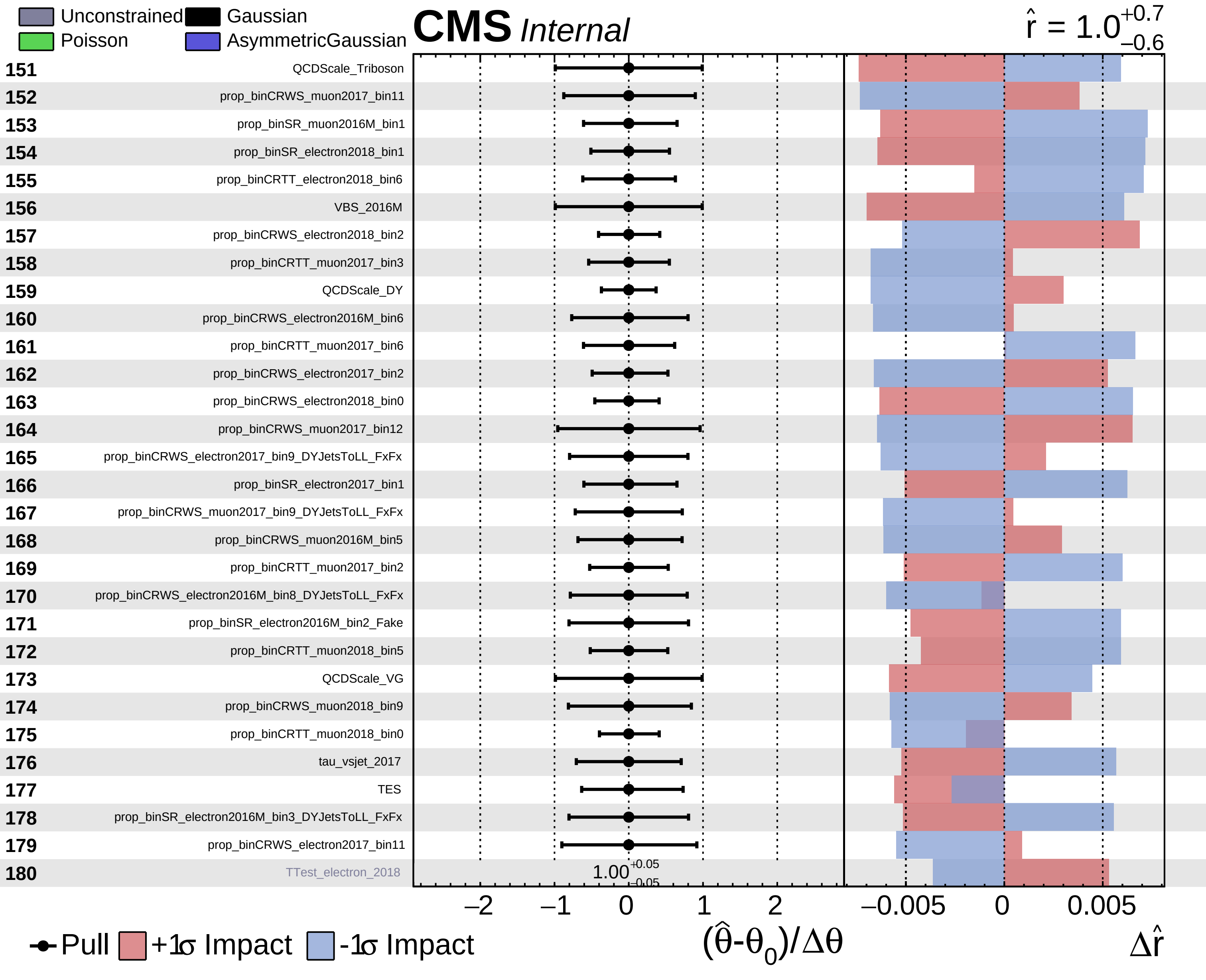


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
61 prop_binSR_muon2016M_bin8
62 prop_binCRTT_muon2017_bin0
63 lumi_2017
64 prop_binCRWS_muon2018_bin8
65 prop_binSR_muon2016M_bin7
66 prop_binSR_electron2017_bin9
67 prop_binSR_electron2018_bin4
68 prop_binSR_electron2017_bin5
69 prop_binSR_electron2016M_bin9
70 prop_binSR_muon2017_bin3
71 prop_binCRWS_muon2018_bin10
72 prop_binSR_electron2016M_bin6
73 prop_binSR_muon2018_bin2
74 prop_binSR_muon2016M_bin4
75 prop_binSR_electron2017_bin8
76 prop_binSR_muon2016M_bin3
77 prop_binCRTT_electron2016M_bin0
78 prop_binCRWS_muon2016M_bin6
79 prop_binCRWS_electron2016M_bin2
80 lumi_2016M
81 prop_binCRWS_muon2017_bin5
82 prop_binSR_muon2017_bin2
83 prop_binCRWS_muon2016M_bin0
84 QCDScale_WpWpJJ_QCD
85 prop_binCRWS_electron2018_bin6
86 prop_binCRWS_muon2017_bin6
87 prop_binCRTT_muon2016M_bin1
88 prop_binCRWS_muon2018_bin12
89 prop_binCRWS_muon2018_bin2
90 prop_binSR_electron2017_bin7
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

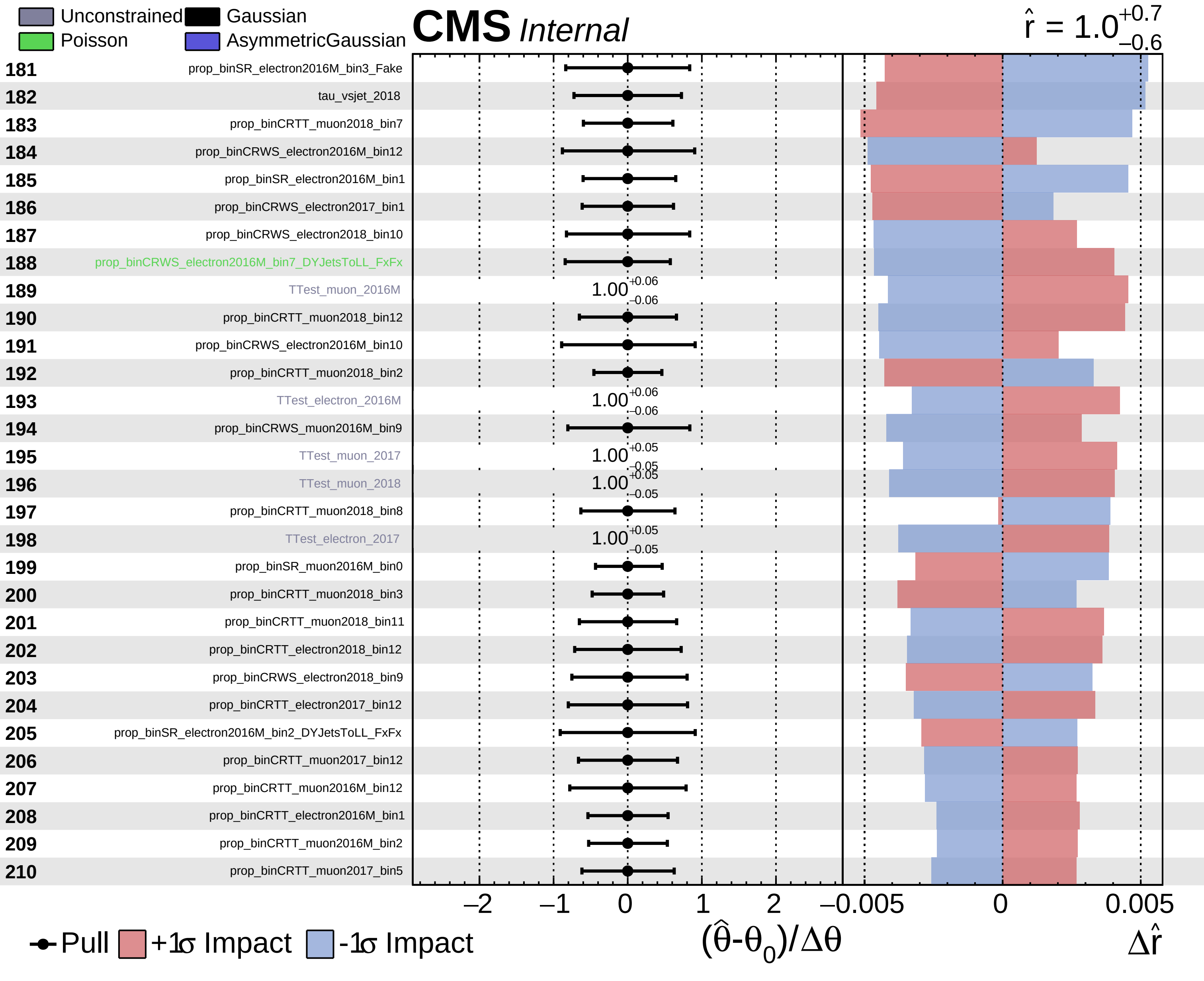


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
91 prop_binSR_electron2017_bin4
92 prop_binSR_electron2016M_bin5
93 prop_binSR_muon2016M_bin2
94 prop_binCRWS_muon2018_bin11
95 prop_binCRWS_muon2016M_bin1
96 prop_binCRWS_muon2018_bin7
97 prop_binSR_electron2016M_bin8
98 prop_binCRWS_muon2018_bin0
99 prop_binCRWS_muon2018_bin5
100 prop_binCRWS_electron2017_bin10
101 pu
102 prop_binCRTT_electron2018_bin2
103 prop_binCRWS_muon2017_bin1
104 jer_2018
105 btag
106 prop_binCRTT_muon2016M_bin0
107 prop_binCRWS_muon2016M_bin7
108 prop_binSR_muon2017_bin1
109 mistag
110 prop_binSR_electron2018_bin3
111 prop_binSR_electron2016M_bin7
112 prop_binCRWS_muon2016M_bin8_DYJetsToLL_FxFx
113 QCDScale_TTdilep
114 prop_binCRWS_electron2017_bin8
115 prop_binSR_muon2018_bin1
116 prop_binCRWS_electron2016M_bin3
117 prop_binSR_electron2017_bin3
118 prop_binCRWS_muon2017_bin8
119 prop_binCRWS_muon2017_bin7
120 prop_binCRWS_muon2017_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

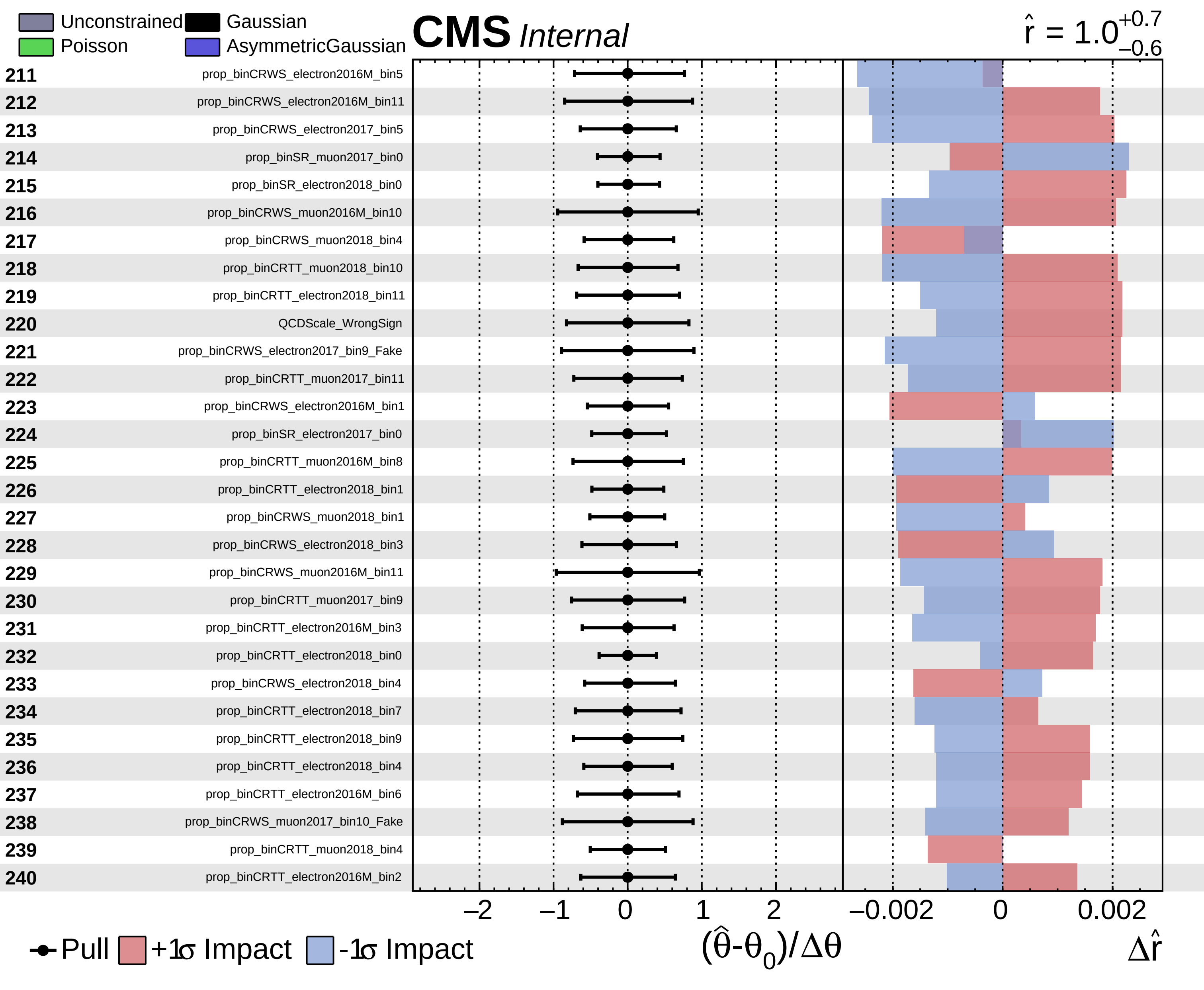

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
121 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
122 prop_binCRWS_electron2018_bin12
123 prop_binCRTT_electron2017_bin1
124 prop_binCRWS_electron2018_bin1
125 prop_binSR_electron2018_bin2
126 prop_binCRTT_muon2018_bin1
127 prop_binCRWS_muon2016M_bin3
128 prop_binCRWS_muon2017_bin2
129 prop_binCRWS_electron2017_bin12
130 prop_binCRWS_electron2018_bin11
131 prop_binCRTT_muon2017_bin1
132 VBS_2018
133 prop_binSR_electron2016M_bin4
134 metUnclust
135 VBS_2017
136 ISR
137 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
138 prop_binCRTT_muon2017_bin10
139 jes
140 prop_binSR_electron2017_bin2
141 prop_binCRWS_electron2018_bin8
142 tau_vsjet_2016M
143 jer_2016M
144 FSR
145 prop_binCRTT_muon2018_bin6
146 QCDScale_TVX
147 prop_binCRWS_electron2017_bin0
148 prop_binCRTT_muon2018_bin9
149 prop_binCRWS_muon2016M_bin12
150 prop_binCRWS_muon2018_bin6
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

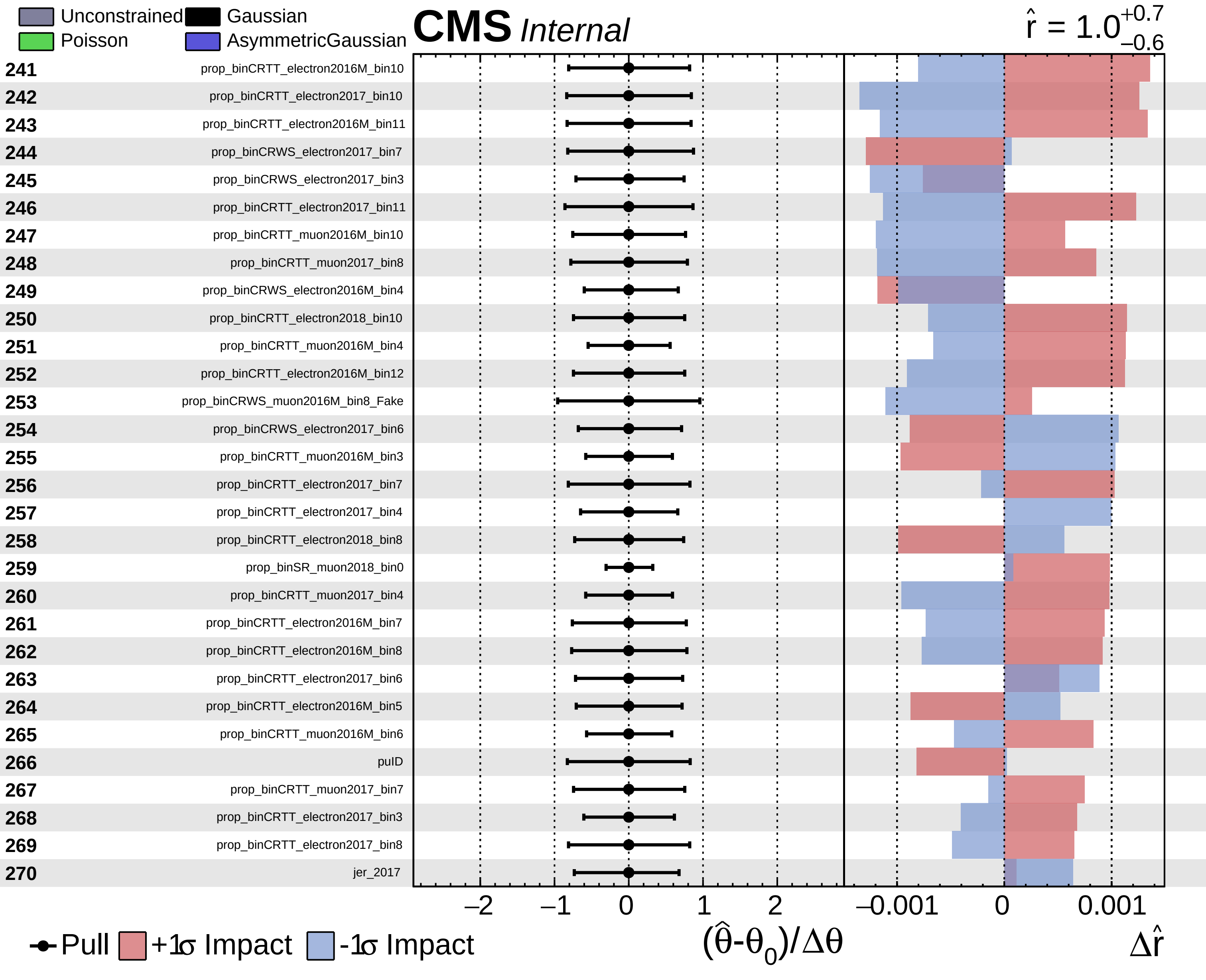


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
151 QCDScale_Triboson
152 prop_binCRWS_muon2017_bin11
153 prop_binSR_muon2016M_bin1
154 prop_binSR_electron2018_bin1
155 prop_binCRTT_electron2018_bin6
156 VBS_2016M
157 prop_binCRWS_electron2018_bin2
158 prop_binCRTT_muon2017_bin3
159 QCDScale_DY
160 prop_binCRWS_electron2016M_bin6
161 prop_binCRTT_muon2017_bin6
162 prop_binCRWS_electron2017_bin2
163 prop_binCRWS_electron2018_bin0
164 prop_binCRWS_muon2017_bin12
165 prop_binCRWS_electron2017_bin9_DYJetsToLL_FxFx
166 prop_binSR_electron2017_bin1
167 prop_binCRWS_muon2017_bin9_DYJetsToLL_FxFx
168 prop_binCRWS_muon2016M_bin5
169 prop_binCRTT_muon2017_bin2
170 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
171 prop_binSR_electron2016M_bin2_Fake
172 prop_binCRTT_muon2018_bin5
173 QCDScale_VG
174 prop_binCRWS_muon2018_bin9
175 prop_binCRTT_muon2018_bin0
176 tau_vsjet_2017
177 TES
178 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
179 prop_binCRWS_electron2017_bin11
180 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

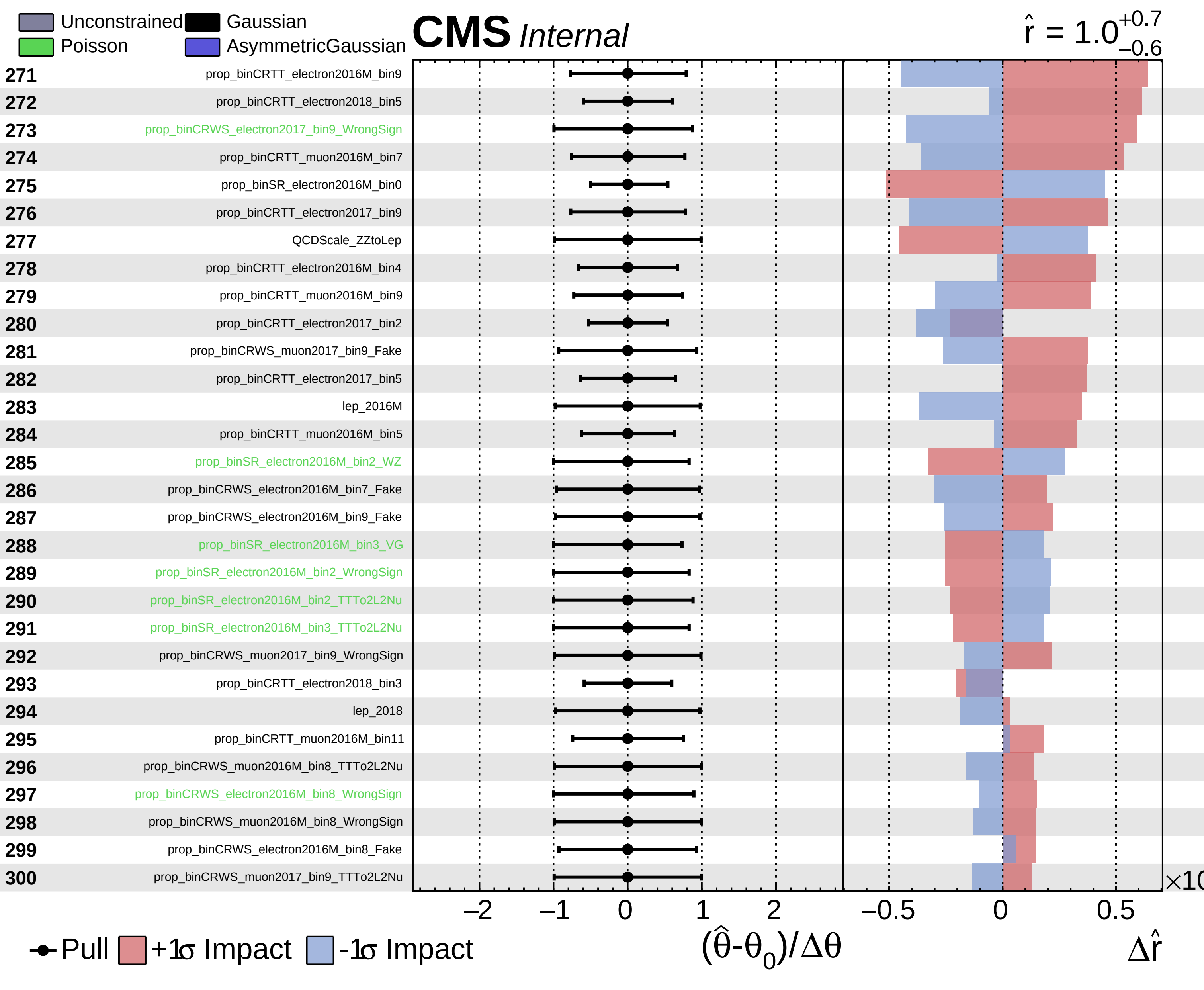


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
181 prop_binSR_electron2016M_bin3_Fake
182 tau_vsjet_2018
183 prop_binCRTT_muon2018_bin7
184 prop_binCRWS_electron2016M_bin12
185 prop_binSR_electron2016M_bin1
186 prop_binCRWS_electron2017_bin1
187 prop_binCRWS_electron2018_bin10
188 prop_binCRWS_electron2016M_bin7_DYJetsToLL_FxFx
189 TTest_muon_2016M $1.00^{+0.06}_{-0.06}$
190 prop_binCRTT_muon2018_bin12
191 prop_binCRWS_electron2016M_bin10
192 prop_binCRTT_muon2018_bin2
193 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
194 prop_binCRWS_muon2016M_bin9
195 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
196 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
197 prop_binCRTT_muon2018_bin8
198 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
199 prop_binSR_muon2016M_bin0
200 prop_binCRTT_muon2018_bin3
201 prop_binCRTT_muon2018_bin11
202 prop_binCRTT_electron2018_bin12
203 prop_binCRWS_electron2018_bin9
204 prop_binCRTT_electron2017_bin12
205 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
206 prop_binCRTT_muon2017_bin12
207 prop_binCRTT_muon2016M_bin12
208 prop_binCRTT_electron2016M_bin1
209 prop_binCRTT_muon2016M_bin2
210 prop_binCRTT_muon2017_bin5
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

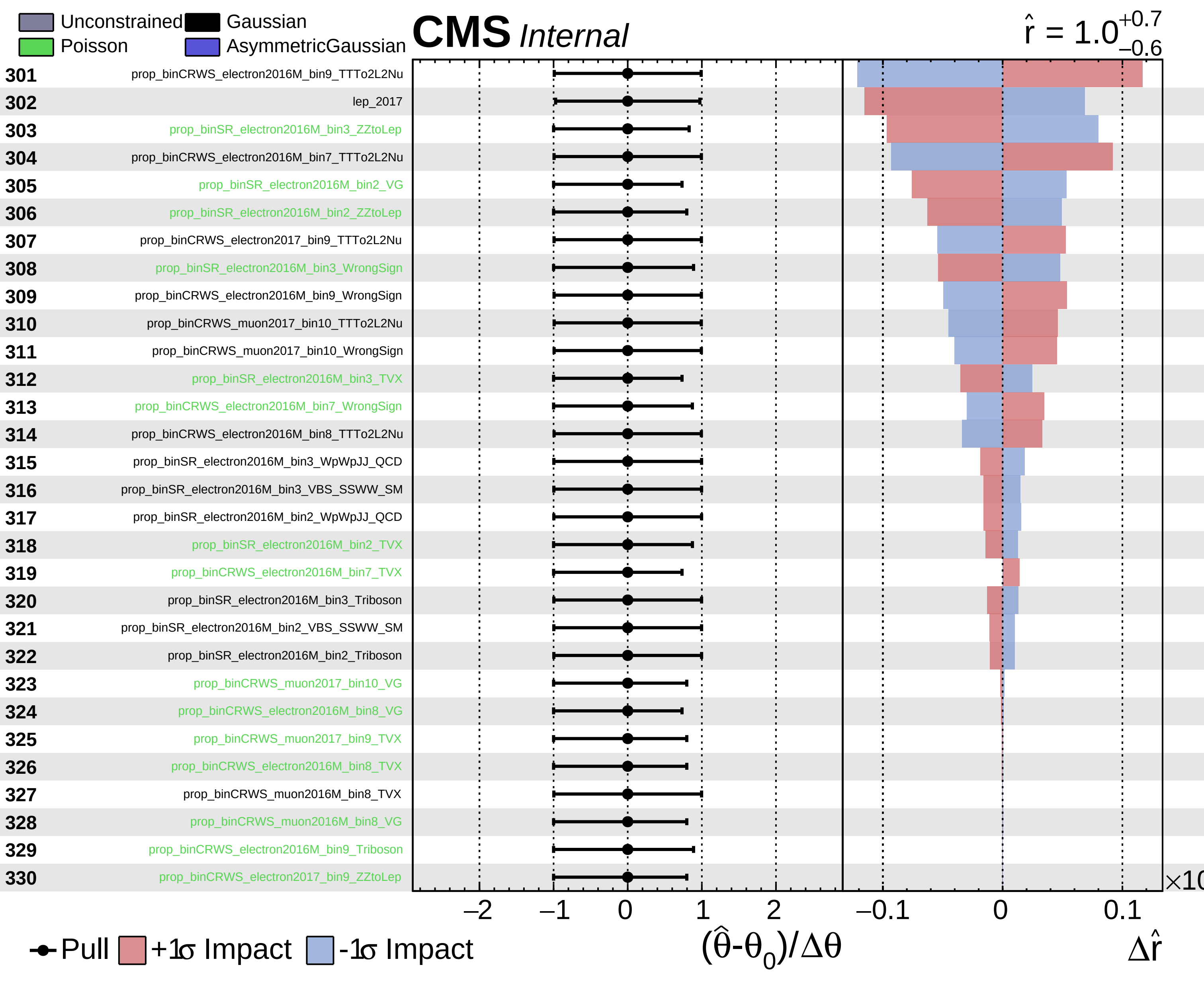

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
211 prop_binCRWS_electron2016M_bin5
212 prop_binCRWS_electron2016M_bin11
213 prop_binCRWS_electron2017_bin5
214 prop_binSR_muon2017_bin0
215 prop_binSR_electron2018_bin0
216 prop_binCRWS_muon2016M_bin10
217 prop_binCRWS_muon2018_bin4
218 prop_binCRTT_muon2018_bin10
219 prop_binCRTT_electron2018_bin11
220 QCDScale_WrongSign
221 prop_binCRWS_electron2017_bin9_Fake
222 prop_binCRTT_muon2017_bin11
223 prop_binCRWS_electron2016M_bin1
224 prop_binSR_electron2017_bin0
225 prop_binCRTT_muon2016M_bin8
226 prop_binCRTT_electron2018_bin1
227 prop_binCRWS_muon2018_bin1
228 prop_binCRWS_electron2018_bin3
229 prop_binCRWS_muon2016M_bin11
230 prop_binCRTT_muon2017_bin9
231 prop_binCRTT_electron2016M_bin3
232 prop_binCRTT_electron2018_bin0
233 prop_binCRWS_electron2018_bin4
234 prop_binCRTT_electron2018_bin7
235 prop_binCRTT_electron2018_bin9
236 prop_binCRTT_electron2018_bin4
237 prop_binCRTT_electron2016M_bin6
238 prop_binCRWS_muon2017_bin10_Fake
239 prop_binCRTT_muon2018_bin4
240 prop_binCRTT_electron2016M_bin2
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

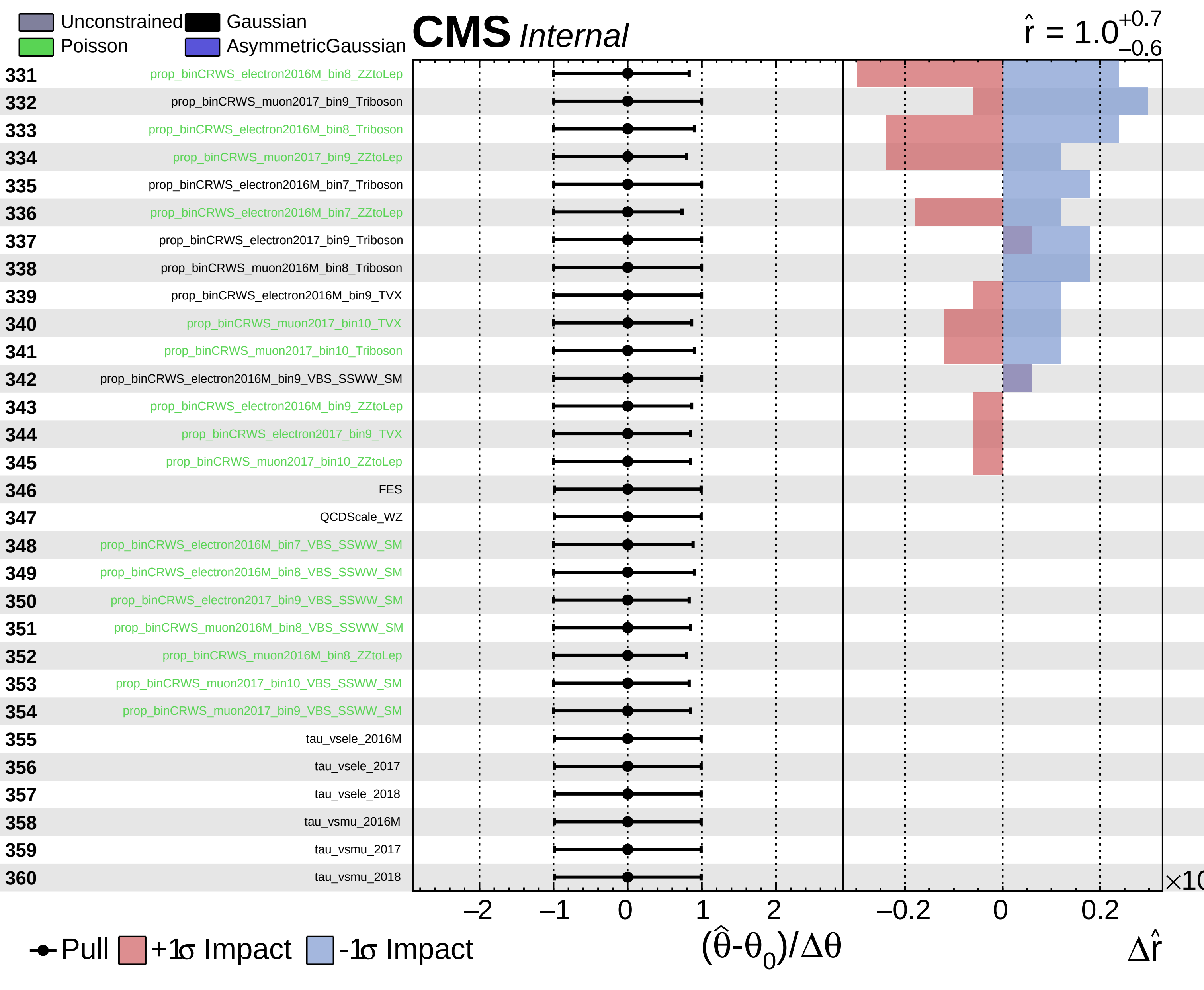


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
241 prop_binCRTT_electron2016M_bin10
242 prop_binCRTT_electron2017_bin10
243 prop_binCRTT_electron2016M_bin11
244 prop_binCRWS_electron2017_bin7
245 prop_binCRWS_electron2017_bin3
246 prop_binCRTT_electron2017_bin11
247 prop_binCRTT_muon2016M_bin10
248 prop_binCRTT_muon2017_bin8
249 prop_binCRWS_electron2016M_bin4
250 prop_binCRTT_electron2018_bin10
251 prop_binCRTT_muon2016M_bin4
252 prop_binCRTT_electron2016M_bin12
253 prop_binCRWS_muon2016M_bin8_Fake
254 prop_binCRWS_electron2017_bin6
255 prop_binCRTT_muon2016M_bin3
256 prop_binCRTT_electron2017_bin7
257 prop_binCRTT_electron2017_bin4
258 prop_binCRTT_electron2018_bin8
259 prop_binSR_muon2018_bin0
260 prop_binCRTT_muon2017_bin4
261 prop_binCRTT_electron2016M_bin7
262 prop_binCRTT_electron2016M_bin8
263 prop_binCRTT_electron2017_bin6
264 prop_binCRTT_electron2016M_bin5
265 prop_binCRTT_muon2016M_bin6
266 puID
267 prop_binCRTT_muon2017_bin7
268 prop_binCRTT_electron2017_bin3
269 prop_binCRTT_electron2017_bin8
270 jer_2017
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
271 prop_binCRTT_electron2016M_bin9
272 prop_binCRTT_electron2018_bin5
273 prop_binCRWS_electron2017_bin9_WrongSign
274 prop_binCRTT_muon2016M_bin7
275 prop_binSR_electron2016M_bin0
276 prop_binCRTT_electron2017_bin9
277 QCDScale_ZZtoLep
278 prop_binCRTT_electron2016M_bin4
279 prop_binCRTT_muon2016M_bin9
280 prop_binCRTT_electron2017_bin2
281 prop_binCRWS_muon2017_bin9_Fake
282 prop_binCRTT_electron2017_bin5
283 lep_2016M
284 prop_binCRTT_muon2016M_bin5
285 prop_binSR_electron2016M_bin2_WZ
286 prop_binCRWS_electron2016M_bin7_Fake
287 prop_binCRWS_electron2016M_bin9_Fake
288 prop_binSR_electron2016M_bin3_VG
289 prop_binSR_electron2016M_bin2_WrongSign
290 prop_binSR_electron2016M_bin2_TTTo2L2Nu
291 prop_binSR_electron2016M_bin3_TTTo2L2Nu
292 prop_binCRWS_muon2017_bin9_WrongSign
293 prop_binCRTT_electron2018_bin3
294 lep_2018
295 prop_binCRTT_muon2016M_bin11
296 prop_binCRWS_muon2016M_bin8_TTTo2L2Nu
297 prop_binCRWS_electron2016M_bin8_WrongSign
298 prop_binCRWS_muon2016M_bin8_WrongSign
299 prop_binCRWS_electron2016M_bin8_Fake
300 prop_binCRWS_muon2017_bin9_TTTo2L2Nu
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
301 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
302 lep_2017
303 prop_binSR_electron2016M_bin3_ZZtoLep
304 prop_binCRWS_electron2016M_bin7_TTTo2L2Nu
305 prop_binSR_electron2016M_bin2_VG
306 prop_binSR_electron2016M_bin2_ZZtoLep
307 prop_binCRWS_electron2017_bin9_TTTo2L2Nu
308 prop_binSR_electron2016M_bin3_WrongSign
309 prop_binCRWS_electron2016M_bin9_WrongSign
310 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
311 prop_binCRWS_muon2017_bin10_WrongSign
312 prop_binSR_electron2016M_bin3_TVX
313 prop_binCRWS_electron2016M_bin7_WrongSign
314 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
315 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
316 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
317 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
318 prop_binSR_electron2016M_bin2_TVX
319 prop_binCRWS_electron2016M_bin7_TVX
320 prop_binSR_electron2016M_bin3_Triboson
321 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
322 prop_binSR_electron2016M_bin2_Triboson
323 prop_binCRWS_muon2017_bin10_VG
324 prop_binCRWS_electron2016M_bin8_VG
325 prop_binCRWS_muon2017_bin9_TVX
326 prop_binCRWS_electron2016M_bin8_TVX
327 prop_binCRWS_muon2016M_bin8_TVX
328 prop_binCRWS_muon2016M_bin8_VG
329 prop_binCRWS_electron2016M_bin9_Triboson
330 prop_binCRWS_electron2017_bin9_ZZtoLep
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.7}_{-0.6}$
331 prop_binCRWS_electron2016M_bin8_ZZtoLep
332 prop_binCRWS_muon2017_bin9_Triboson
333 prop_binCRWS_electron2016M_bin8_Triboson
334 prop_binCRWS_muon2017_bin9_ZZtoLep
335 prop_binCRWS_electron2016M_bin7_Triboson
336 prop_binCRWS_electron2016M_bin7_ZZtoLep
337 prop_binCRWS_electron2017_bin9_Triboson
338 prop_binCRWS_muon2016M_bin8_Triboson
339 prop_binCRWS_electron2016M_bin9_TVX
340 prop_binCRWS_muon2017_bin10_TVX
341 prop_binCRWS_muon2017_bin10_Triboson
342 prop_binCRWS_electron2016M_bin9_VBS_SSWW_SM
343 prop_binCRWS_electron2016M_bin9_ZZtoLep
344 prop_binCRWS_electron2017_bin9_TVX
345 prop_binCRWS_muon2017_bin10_ZZtoLep
346 FES
347 QCDScale_WZ
348 prop_binCRWS_electron2016M_bin7_VBS_SSWW_SM
349 prop_binCRWS_electron2016M_bin8_VBS_SSWW_SM
350 prop_binCRWS_electron2017_bin9_VBS_SSWW_SM
351 prop_binCRWS_muon2016M_bin8_VBS_SSWW_SM
352 prop_binCRWS_muon2016M_bin8_ZZtoLep
353 prop_binCRWS_muon2017_bin10_VBS_SSWW_SM
354 prop_binCRWS_muon2017_bin9_VBS_SSWW_SM
355 tau_vsele_2016M
356 tau_vsele_2017
357 tau_vsele_2018
358 tau_vsmu_2016M
359 tau_vsmu_2017
360 tau_vsmu_2018
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$